The Stars

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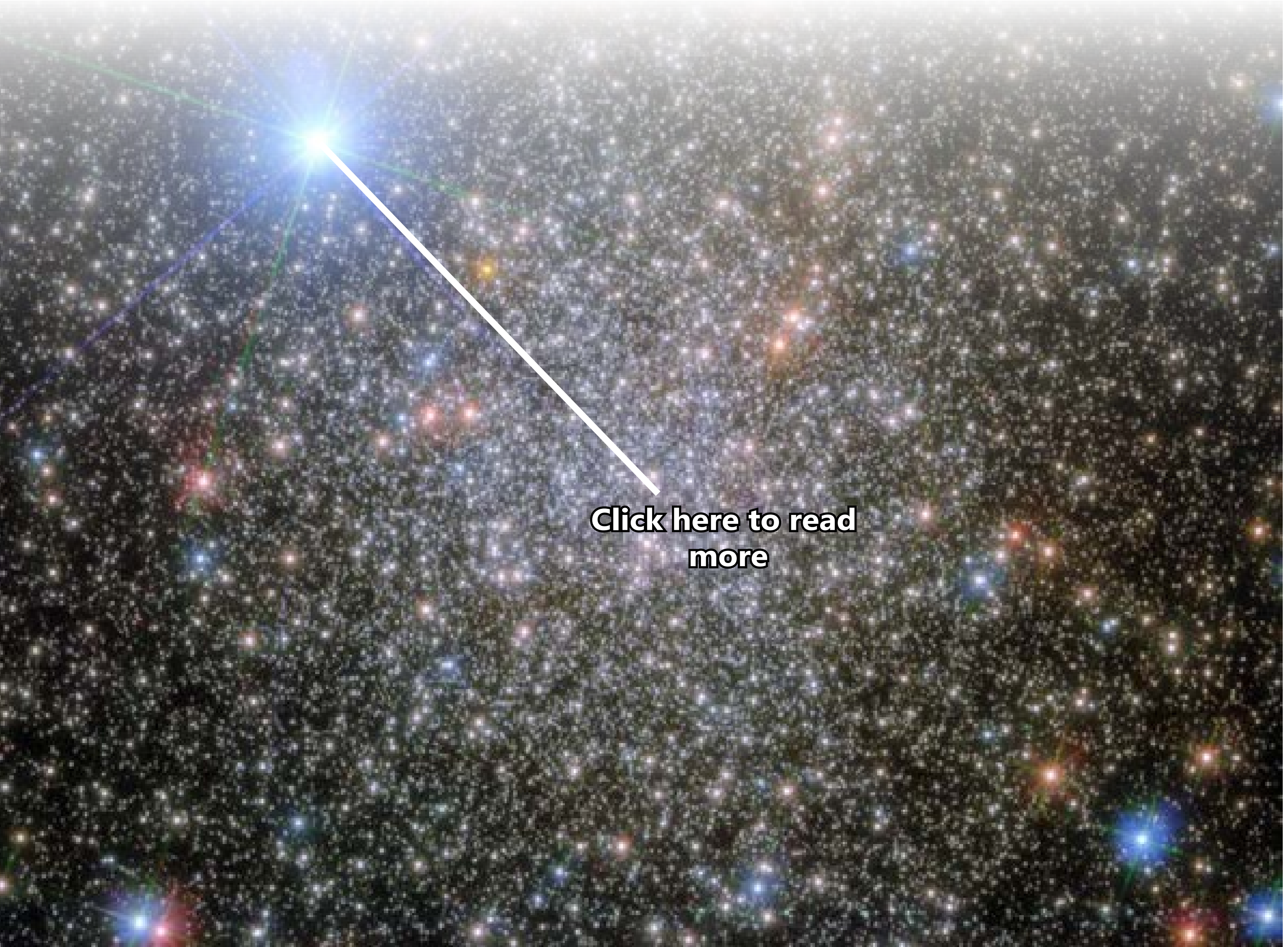
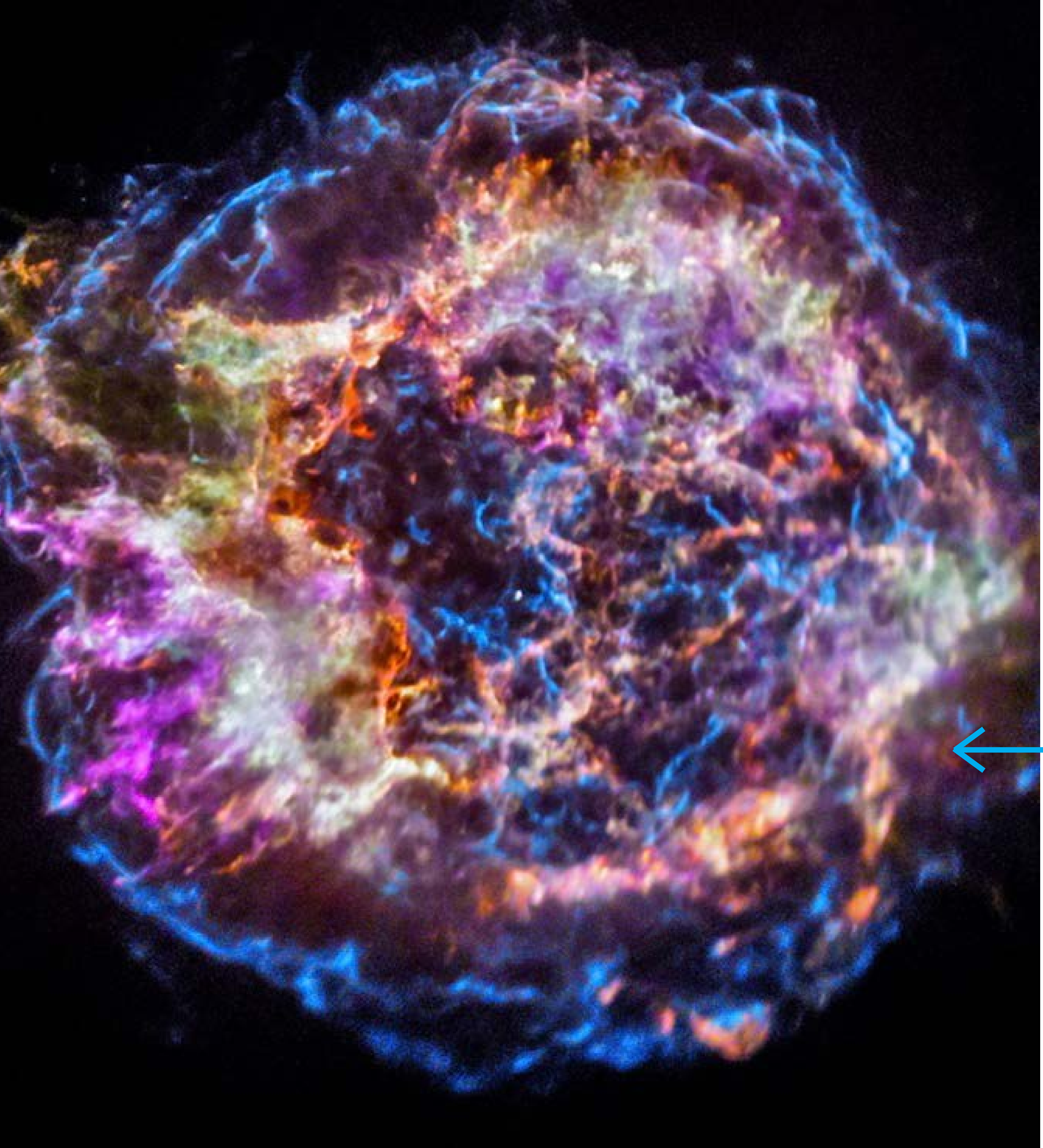
Stars

are giant, luminous spheres of plasma scattered throughout the universe, including billions within the Milky Way galaxy and countless more in other galaxies. A star forms from a massive cloud of gas, primarily hydrogen and helium, that collapses under its gravitational pull and eventually ignites due to nuclear fusion in its core, converting mass into energy.



The life cycles of stars

are dictated by their initial mass, leading to various outcomes such as the formation of intermediate-mass stars like the sun, high-mass stars, and low-mass stars. Intermediate-mass stars, for instance, evolve from a cloud into a protostar, eventually becoming main-sequence stars that fuse hydrogen in their cores.



High-mass stars

follow a quicker evolution, with some forming and dying in just tens of thousands of years. These stars, when they exhaust their nuclear fuel, may explode in a supernova, leaving behind remnants such as neutron stars or black holes.

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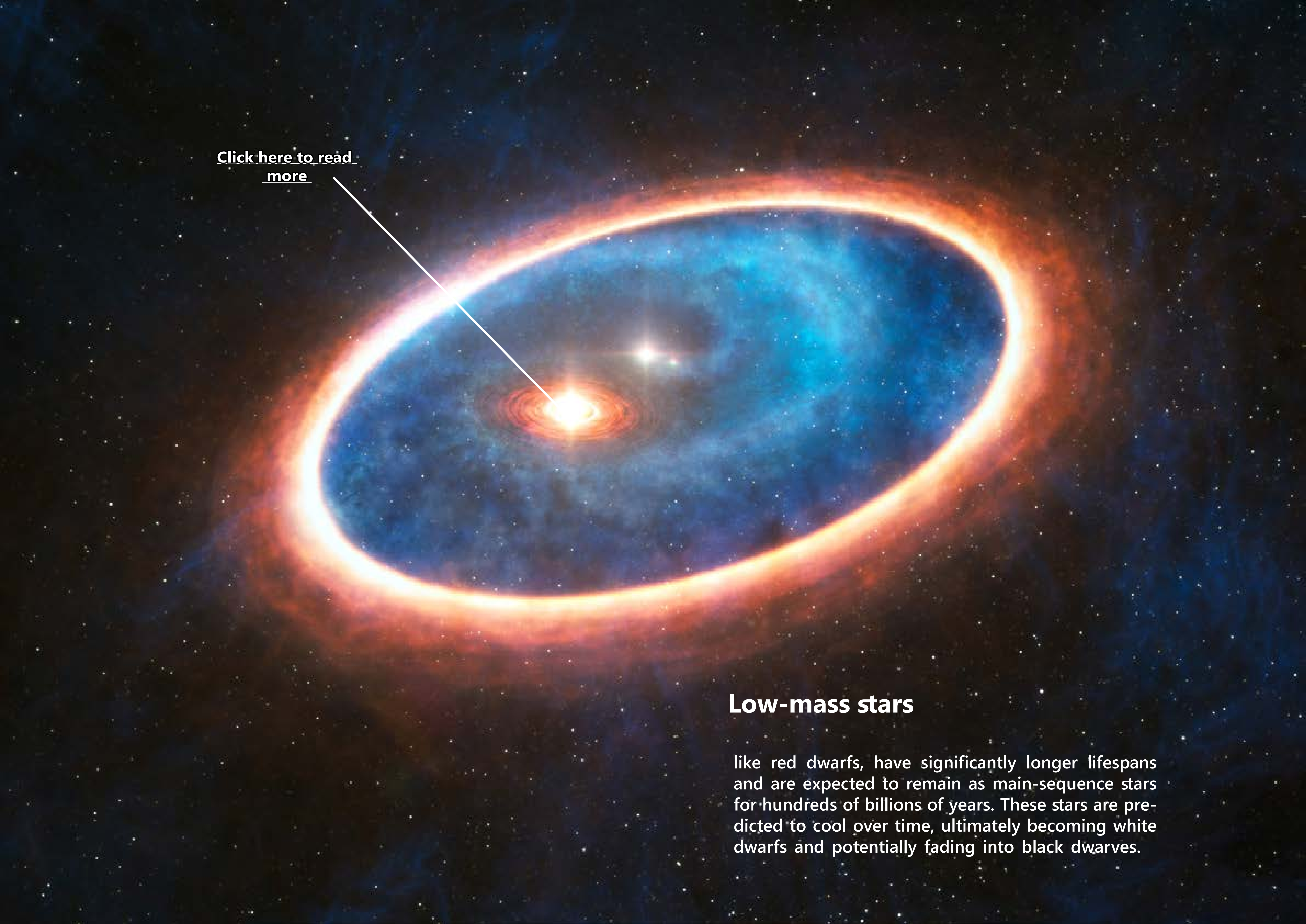
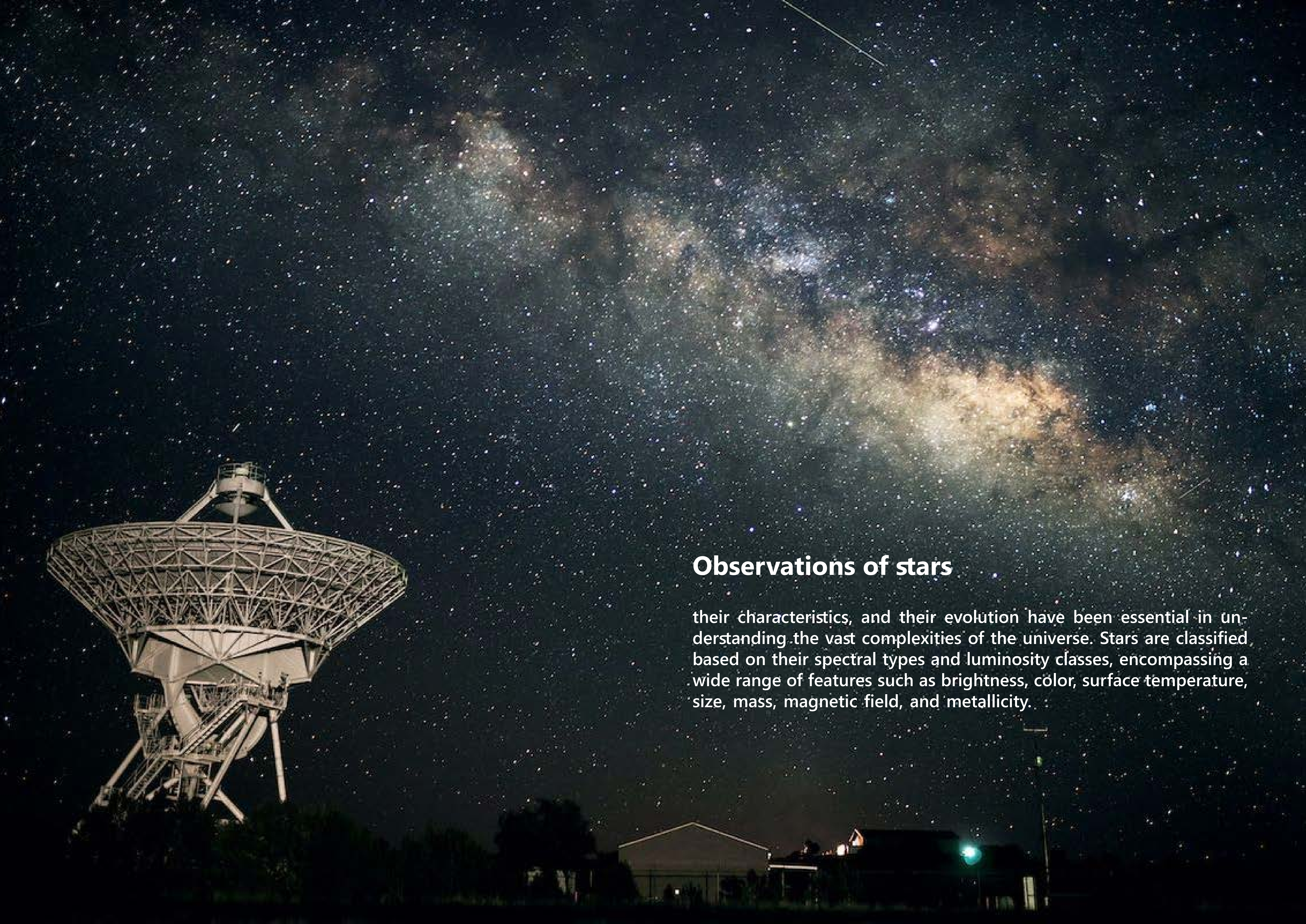


Low-mass stars

like red dwarfs, have significantly longer lifespans and are expected to remain as main-sequence stars for hundreds of billions of years. These stars are predicted to cool over time, ultimately becoming white dwarfs and potentially fading into black dwarves.



Observations of stars

their characteristics, and their evolution have been essential in understanding the vast complexities of the universe. Stars are classified based on their spectral types and luminosity classes, encompassing a wide range of features such as brightness, color, surface temperature, size, mass, magnetic field, and metallicity.

Astronomical advancements

including the invention of the telescope and the development of various observing technologies, have significantly contributed to expanding our knowledge of stars. Technologies like radio telescopes, infrared telescopes, and space-based observatories like the Hubble Space Telescope have revolutionized our understanding of stars and their evolution.

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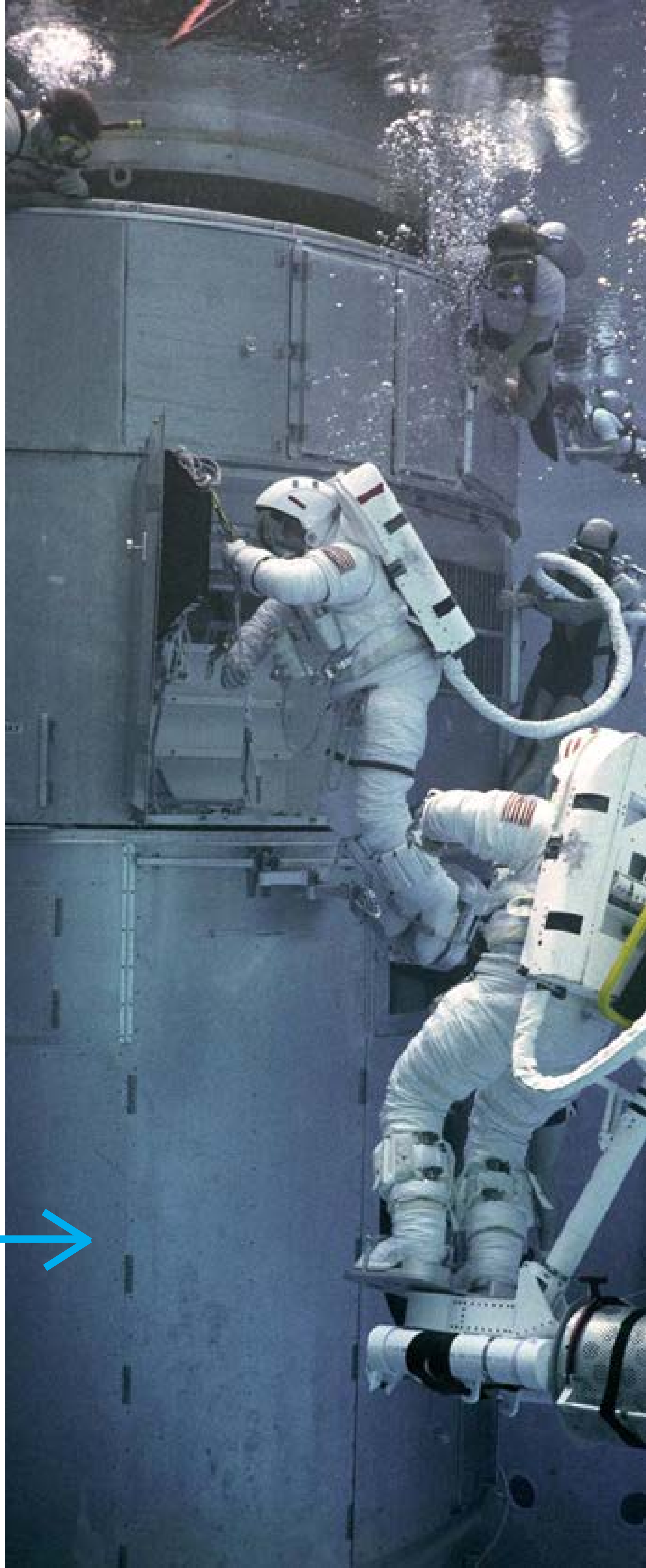
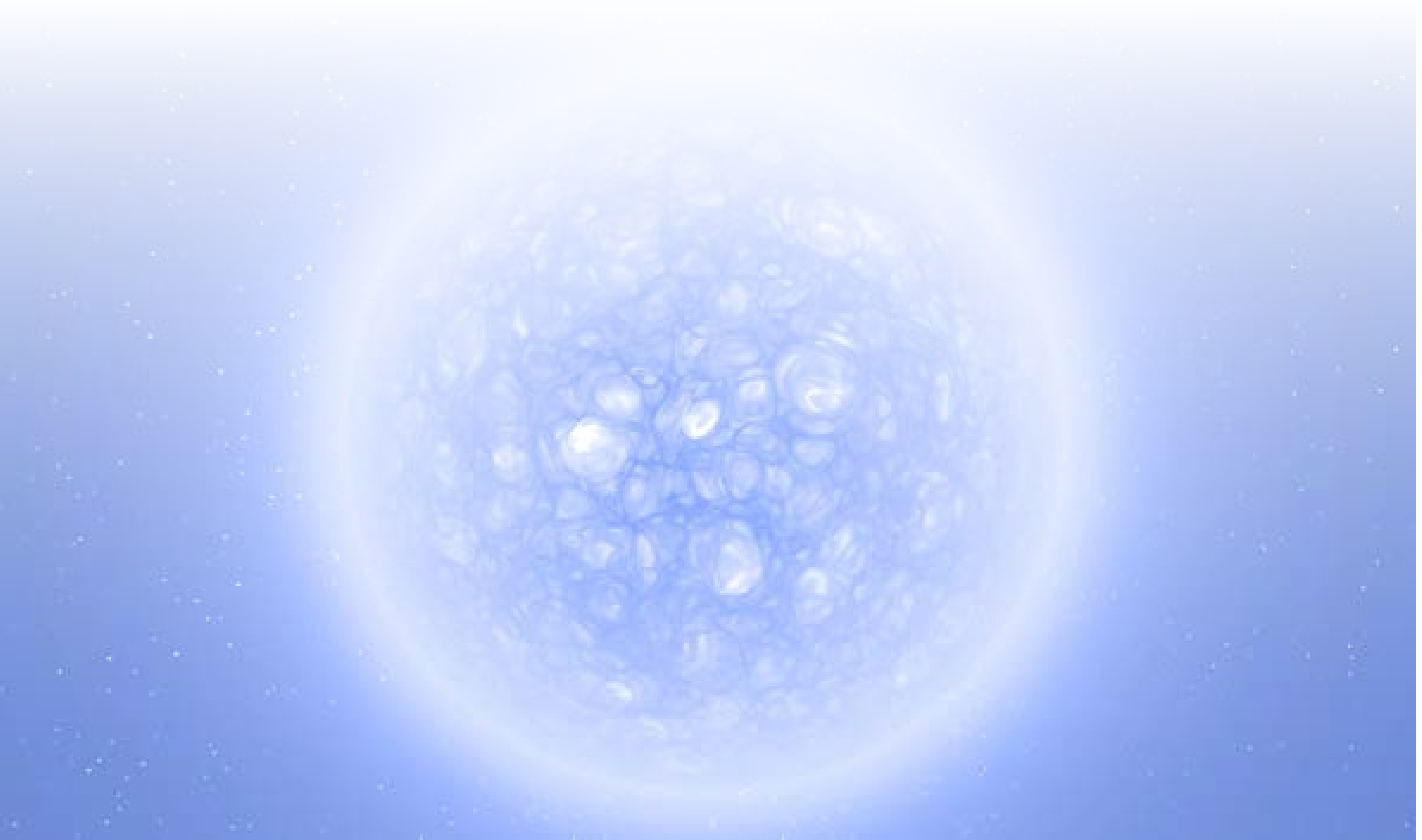


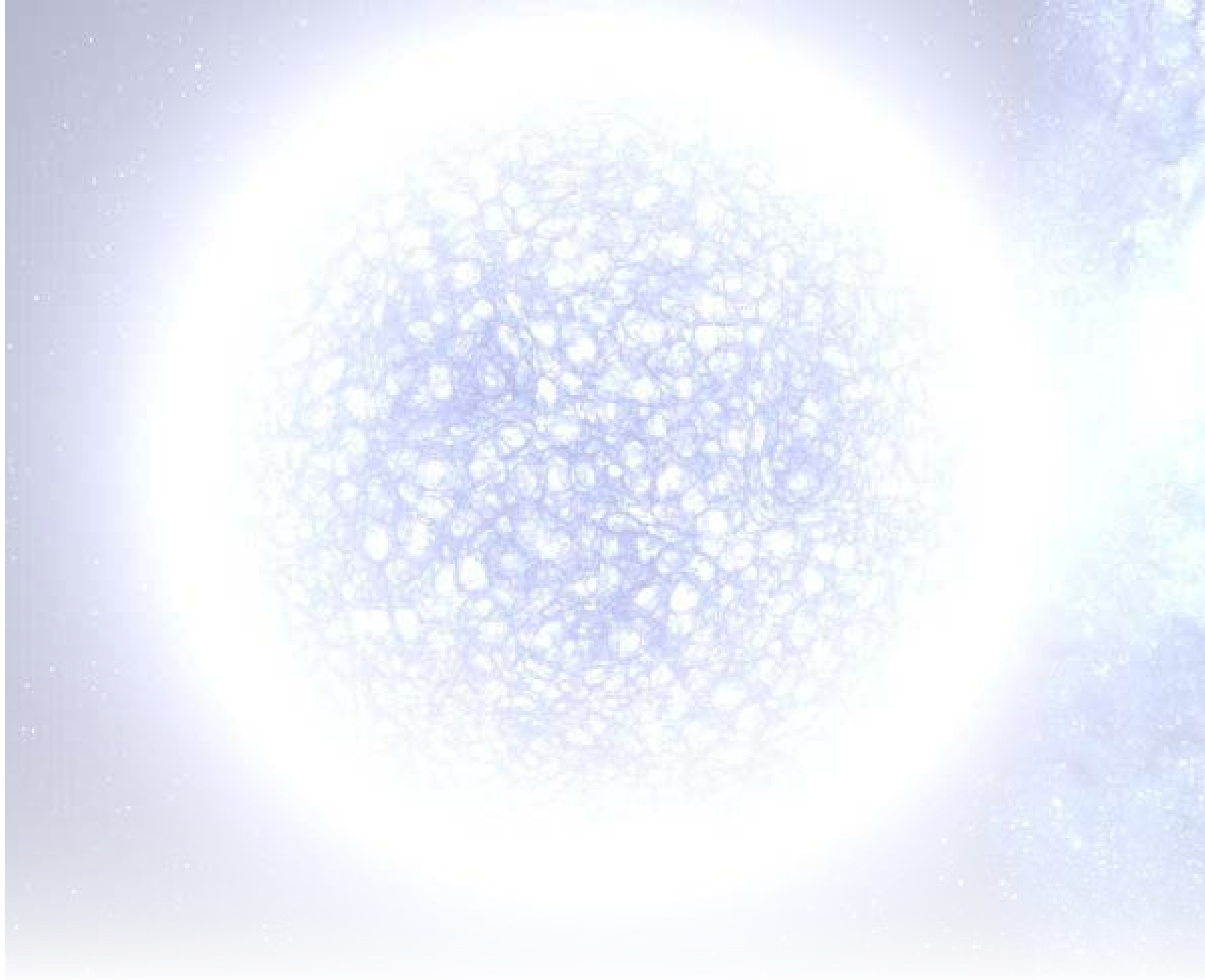
- What are the 7 main types of stars ?

The seven main types of stars are classified based on a system called spectral classification, grouping stars by their temperature, color, and brightness. These groups are denoted by the letters O, B, A, F, G, K, and M, with each letter representing a different type of star.

1. O-class stars:

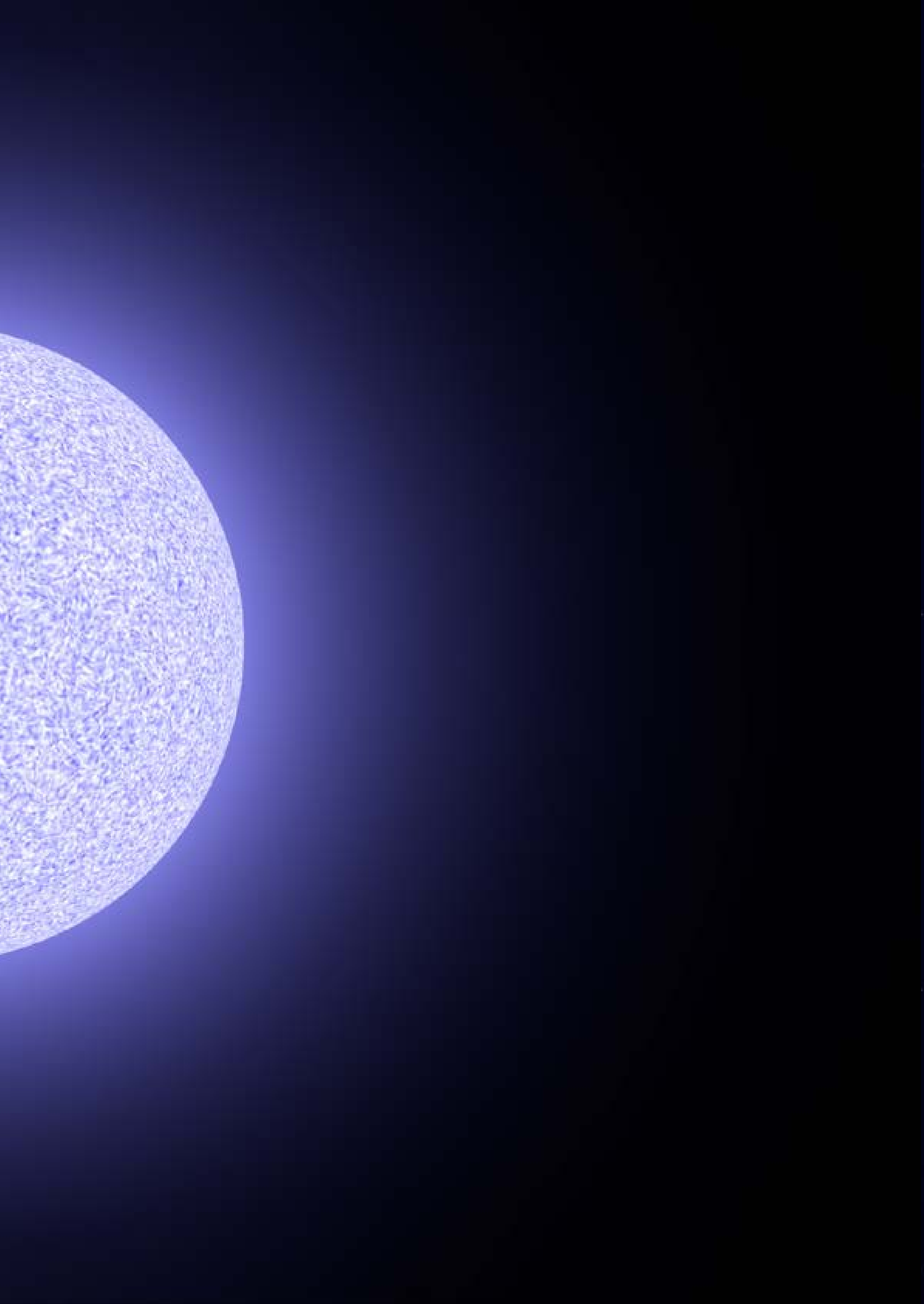
O-type stars are among the hottest and most luminous stars with the majority of their emitted light in the ultraviolet range. They are rare, comprising only about 0.00003% of main- sequence stars in the solar neighborhood.





2. A-class stars:

A-type stars are white or bluish-white with strong hydrogen lines and ionized metal lines. They are among the more common naked-eye stars and account for about 0.625% of main-sequence stars in the solar neighborhood.



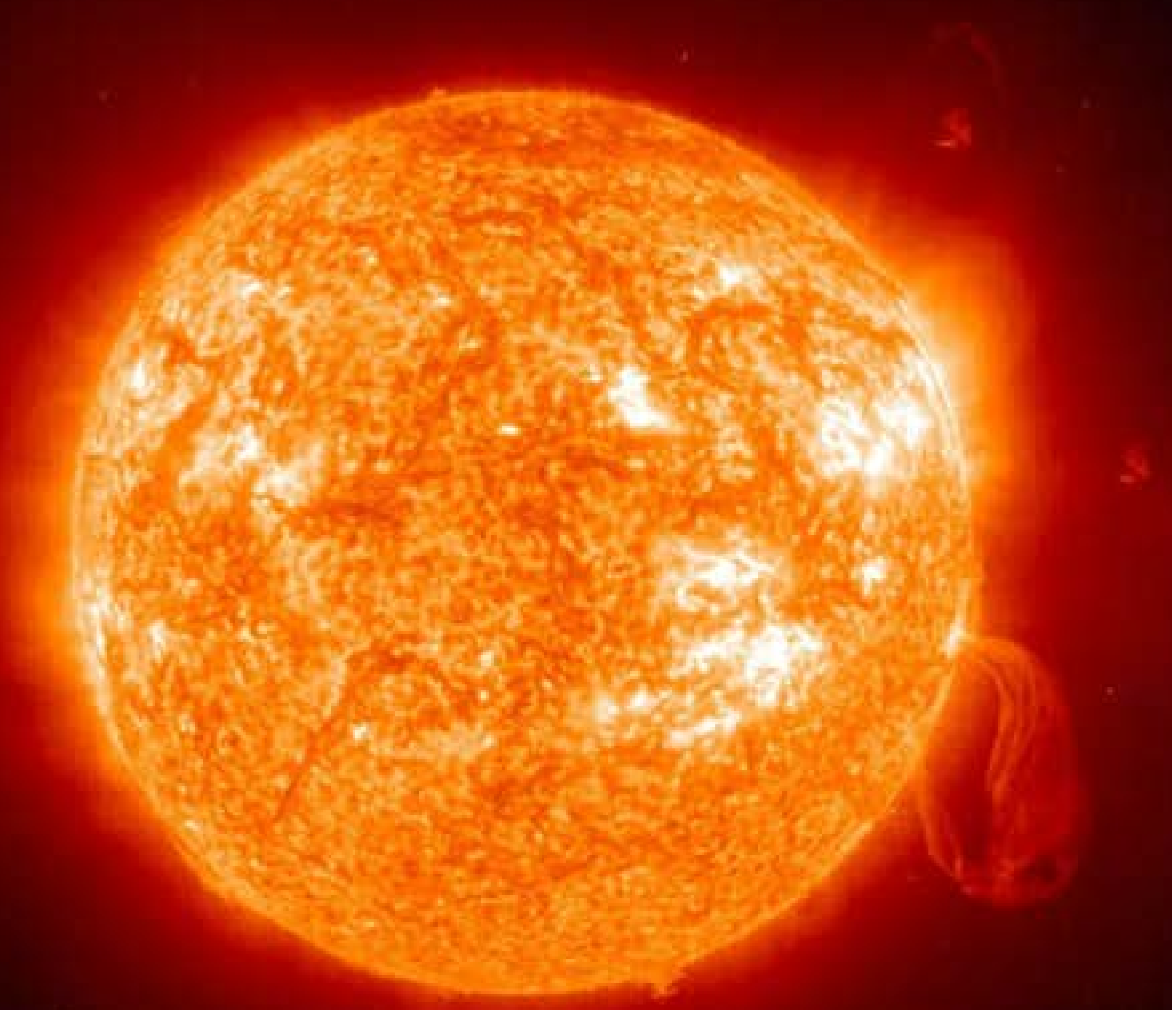
3. B-class stars:

B-type stars are luminous and blue, characterized by neutral helium lines and moderate hydrogen lines. They have relatively short lifespans due to their high energy output. B-type stars are relatively uncommon and are found in OB associations within giant molecular clouds.

4. F-class stars:

F-type stars exhibit strengthening spectral lines of calcium II and are characterized by weaker hydrogen lines and ionized metals. They have a white color and make up about 3.03% of main-sequence stars in the solar neighvvborhood.

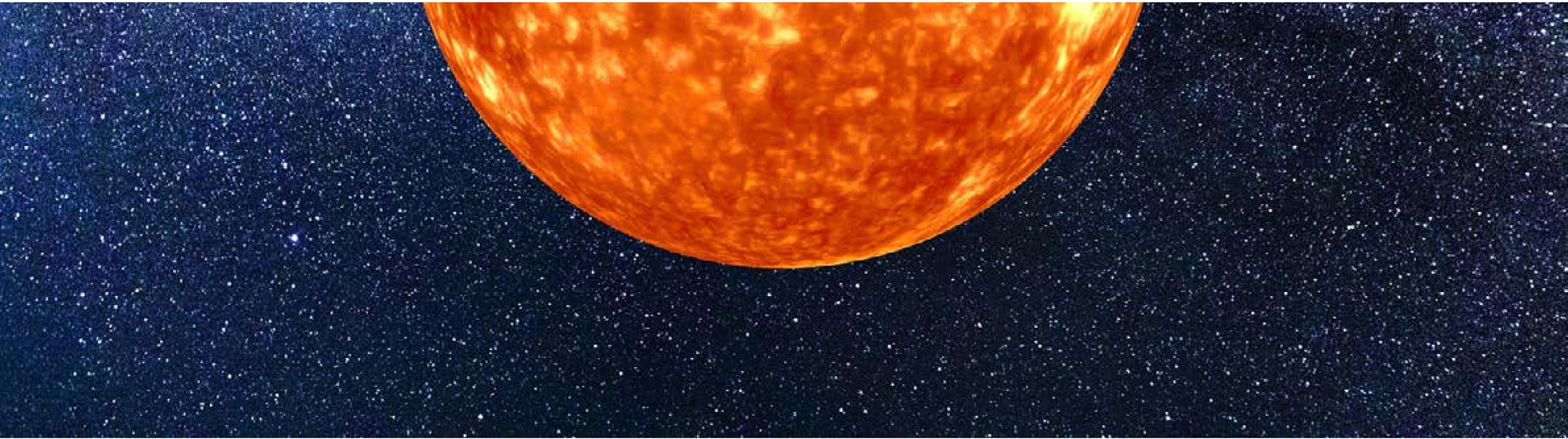
5. G-class stars:

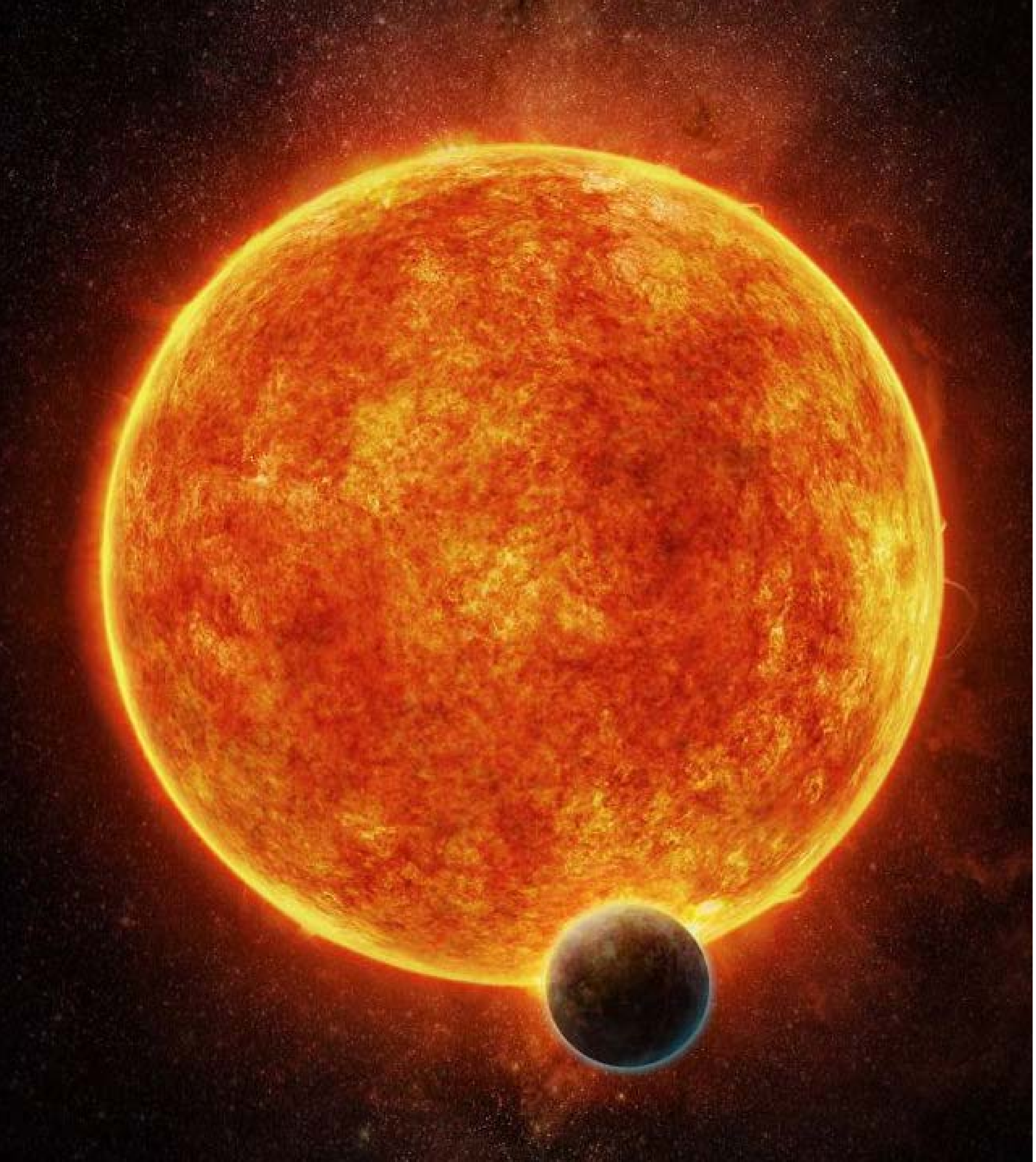


G-type stars, including the Sun, showcase prominent spectral lines of calcium II and have weaker hydrogen lines compared to F-type stars. They also have neutral metals and are known for a visible spike in CN (cyanide) molecule bands. G-type stars constitute approximately 7.5% of main-sequence stars in the solar neighborhood.

6. K-class stars:

K-type stars have light orange to pale yellowish-orange colors. They exhibit very weak hydrogen lines and neutral metal lines. These stars account for about 12% of main-sequence stars in the solar neighborhood.





7. M-class stars:

M-type stars are orangish-red and have very weak hydrogen lines and extremely low luminosity. They are the most common type of star, making up approximately 76% of main-sequence stars in the solar neighborhood.

Each of these star types plays a crucial role in the stellar population, contributing to the vast diversity and understanding of the universe

Stars twinkling in the night sky tell a tale of cosmic evolution and celestial harmony. Their eternal shimmer acts as a prompt of the immensity and magnificence of the cosmos, Their timeless glow serves as a reminder of the vastness and beauty of the universe, sparking contemplation and inspiration in the hearts of stargazers.